Canvas #02/10

Canvases:

ETHICAL CONSIDERATIONS

SOCIAL ROBOT CO-DESIGN CANVASES

Consider potential ethical problems, and potential solutions –both from the user's and robot's perspectives. Consider the boxes to be guidelines: you don't need to fill each one.

Physical safety

Machines can pinch or crush the user. How is this mitigated?

	PROBLEM	SOLUTION
USER	A freestanding kiosk near vulnerable patients could obstruct movement or block emergency pathways.	Staff position the kiosk away from clinical pathways and busy walkways.
ROBOT		The kiosk maintains a safe, comfortable distance from patients, with rounded edges and stable, hospital-grade construction.

Data security

Is the robot in a unique data collection position? How is the user's data protected?

	PROBLEM	SOLUTION
USER	Sensitive patient information could be exposed through a public-facing, walk-up kiosk.	Patients enter only the information requested and step away once check-in is complete.
ROBOT		Displays only the minimum necessary information, encrypts all transmitted data, and securely sends records to the EHR.

Transparency

How does the robot share an accurate perception of its abilities, intentions and constraints, so the user can evaluate their trust in it?

	PROBLEM	SOLUTION
USER	Patients may not understand what an AI-assisted kiosk does or its limitations, risking unrealistic expectations.	The patient reads or listens to the on-screen introduction before beginning.
ROBOT		Clearly states its purpose and limits, and confirms the AI supports rather than replaces clinical staff (informed consent).

Equality across users

Robots' algorithms can be biased. A robot's appearance could reinforce harmful stereotypes. What are potential issues?

	PROBLEM	SOLUTION
USER	Patients differ in age, language, literacy and physical ability, risking unequal access to self-check-in.	Patients select the language, text size, or voice-guidance option that suits their needs.
ROBOT		Provides multilingual support, voice guidance, adjustable text size and wheelchair-accessible design.

Emotional consideration

People have been shown to form emotional attachments to robots, as if they were alive. Is this a potential problem?

	PROBLEM	SOLUTION
USER	Patients may feel anxious interacting with an unfamiliar automated system, especially if unwell.	Patients can request human help from staff at any point in the interaction.
ROBOT		Uses friendly visual cues and a calm, reassuring tone, with an obvious button to request staff assistance.

Behaviour enforcement

People could transfer their inappropriate behaviour, such as rudeness, from robots to humans. How is this mitigated?

	PROBLEM	SOLUTION
USER	A patient identified as high risk, or unable to complete the interaction, might not receive timely care.	Staff respond promptly whenever the kiosk raises an escalation alert.
ROBOT		Automatically and immediately refers high-risk, incomplete, or abnormal cases to a triage nurse.

Cite as: Axelsson, M., Oliveira, R., Racca, M., & Kyrki, V. (2021). Social robot co-design canvases: A participatory design framework. ACM Transactions on Human-Robot Interaction (THRI), 11(1), 1-39.

THRI paper: